

Project Report- Hangman Game

1. Introduction

The Hangman Game is a classic word guessing game where the player tries to guess a hidden word by suggesting one letter at a time.

In this project, we implement a simple text-based Hangman game using Python.

The project demonstrates key Python concepts such as:

random module for word selection

while loops for repeated gameplay

If-else statements for decision making

strings and **lists** for handling words and guesses

2. Objectives

Create a simple text-based game using **Python**.

Limit incorrect guesses to **6 attempts**.

Use a **small predefined list of 5 words**.

Provide clear console-based input/output.

3. Scope

Simplified Hangman game
(text-only, no graphics or audio).

Works in any terminal/console environment.

Focuses on **basic Python programming concepts**.

4. Code Implementation

Below is the Python code for the Hangman game:

```

import random

def hangman():
    print(" Welcome to Hangman Game ")
    print("Guess the hidden word, one letter at a time.")
    print("You have 6 incorrect attempts. Good luck!\n")

    # Predefined list of words (5 words)
    words = ["python", "school", "apple", "computer", "friend"]

    # Randomly select a word
    secret_word = random.choice(words)
    guessed_letters = []
    attempts_left = 6

    # Create display version of the word
    hidden_word = ["_"] * len(secret_word)

    # Game loop
    while attempts_left > 0 and "_" in hidden_word:
        print("Word: ", " ".join(hidden_word))
        print(f"Attempts left: {attempts_left}")
        print("Guessed letters: ", " ".join(guessed_letters) if guessed_letters else "None")

        # Player input
        guess = input("Enter a letter: ").lower()

        # Validate input
        if len(guess) != 1 or not guess.isalpha():
            print("✗ Please enter a single alphabet letter.\n")
            continue

        if guess in guessed_letters:
            print("⚠ You already guessed that letter.\n")
            continue

        guessed_letters.append(guess)

        # Check if guess is correct
        if guess in secret_word:
            print("✓ Good guess!\n")
            for i in range(len(secret_word)):
                if secret_word[i] == guess:
                    hidden_word[i] = guess
            else:
                print("✗ Wrong guess!\n")
                attempts_left -= 1

    # Game result
    if "_" not in hidden_word:
        print(" Congratulations! You guessed the word:", secret_word)

```

```
else:
    print("Game Over! The word was:", secret_word)

# Run the game
if __name__ == "__main__":
    hangman()
```

5. Example Run (Output Screenshot)

Welcome to Hangman Game
Guess the hidden word, one letter at a time.
You have 6 incorrect attempts. Good luck!

Word: _ _ _ _ _
Attempts left: 6
Guessed letters: None
Enter a letter: a
✓ Good guess!

Word: a _ _ _ _
Attempts left: 6
Guessed letters: a
Enter a letter: z
✗ Wrong guess!

...
Congratulations! You guessed the word: apple

6. Conclusion

This project successfully implements a text-based Hangman game using Python. It demonstrates the use of loops, conditionals, lists, and random word selection in a fun and interactive way.

The game can be extended in the future with:

More words from a file/API

Graphical user interface (GUI)

Multiplayer support

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